Literature Review

MedXVLM — A Vision-Language Model for Chest X-Rays

B.Sc. AI & Data Science — Major Project

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Why this review covers what it covers

Before any code got written, it made sense to look into what people had already done in this space, so this project wouldn't end up badly reinventing something that already exists, and so the choices made later could actually be justified instead of picked randomly. This project (MedXVLM) has three parts — it looks at a chest X-ray and lines it up with its written report, it predicts 14 possible findings from the image, and it eventually tries to write a short report on its own. So the reading was organised around those three things, plus a couple of extra topics: how to make the model explain itself, and how to fine-tune a language model without needing a huge GPU. What follows is grouped roughly by which part of the project each paper is relevant to.

Not every paper below got the same amount of attention — some were read in full, some mostly through the abstract, figures, and results section with the method skimmed. The goal wasn't to produce an exhaustive academic summary of each one, but to understand what each paper is doing and, more importantly, *why it matters for this specific project*, which is what the notes below try to explain rather than summarising each paper in isolation.

1. Papers about classifying chest X-rays (the "old" way)

This is a good place to start because it's basically the baseline this project gets compared against — the "before" picture.

ChestX-ray8 / ChestX-ray14

Wang, Peng, Lu, Lu, Bagheri & Summers, 2017, CVPR

This is the paper that basically started large-scale chest X-ray deep learning. NIH released about 109,000 X-ray images from over 32,000 patients, and instead of paying doctors to label all of them by hand, they

used NLP to pull disease labels out of the existing radiology reports. The original paper had 8 labels, and a later version of the dataset had 14 labels, which is why it gets called both "ChestX-ray8" and "ChestX-ray14" depending on who's citing it — both names refer to the same underlying release. Almost every other paper in this review either trains on this dataset or compares itself against it, which is why it's worth starting here even though it isn't directly what this project uses.

CheXNet

Rajpurkar, Irvin, Zhu et al., 2017, arXiv (not peer-reviewed, just a report)

This one trained a 121-layer DenseNet on the dataset above to detect pneumonia, and later all 14 conditions, and the headline result was that it beat the average of four practicing radiologists on F1 score for pneumonia detection. It's a good example of the "classic" approach: one CNN, straight from pixels to labels, no text involved anywhere. This project's baseline model (built in Phase 2 to compare against) is basically the same idea — a plain image classifier with no text branch — just with a Vision Transformer instead of a DenseNet, since that's more standard now.

CheXpert

Irvin, Rajpurkar, Ko et al., 2019, AAAI

This one matters directly because this project's classifier (Stage 2) predicts the exact same 14 labels that CheXpert defined (things like Cardiomegaly, Edema, Pneumothorax, etc.). CheXpert's actual contribution was a smarter, rule-based text labeler that reads a report and outputs positive/negative/uncertain for each of the 14 conditions, instead of forcing everything into a strict yes/no. A version of this labeler hasn't been built yet for this project — that's coming up next in the timeline — but this paper is the reason the label set looks the way it does.

CheXbert

Smit, Jain, Rajpurkar, Pareek, Ng & Lungren, 2020, EMNLP

This one basically replaced CheXpert's hand-written rules with a trained BERT model that does the same labeling job, but more accurately. It's included here mainly because it's become the standard way people score

how "clinically correct" a generated report is — running CheXbert on both the real report and the AI-generated one and comparing the labels it extracts from each. That will matter later, once Stage 3 (report generation) actually exists and needs a way to be evaluated beyond just "does it sound like English".

2. Papers about connecting images and text (the CLIP idea)

This section is the core of the project, since Stage 1 is exactly this kind of model.

CLIP

Radford, Kim, Hallacy et al., 2021, ICML

CLIP is the paper that basically everything below is a reaction to or a medical version of. The idea is deceptively simple: take a huge pile of (image, caption) pairs from the internet — 400 million of them in CLIP's case — and train an image encoder and a text encoder together so that the embedding of an image ends up close to the embedding of its matching caption, and far from the embeddings of all the other captions in the batch. This is called a contrastive loss (specifically InfoNCE, which takes a bit of work to actually follow mathematically, but the intuition is "pull matching pairs together, push everything else apart"). Once trained, an image can be classified with zero extra training just by comparing it against text prompts like "a photo of a dog" vs "a photo of a cat". Stage 1 of this project uses basically this exact loss, just on chest X-rays and their reports instead of random internet photos and captions.

ConVIRT

Zhang, Jiang, Miura, Manning & Langlotz, arXiv 2020, later published MLHC 2022

This is basically "CLIP but for chest X-rays and their reports", and interestingly it actually came out slightly before CLIP was public. It used a ResNet50 for images and a ClinicalBERT for text, both trained with a two-

way contrastive loss on paired X-rays and reports. Their headline result was that the resulting image encoder needed only about 10% of the labeled data to match a normal supervised ImageNet-pretrained model on downstream classification tasks — which is basically the whole point of doing this kind of pretraining. Stage 1's "global" loss (the image-vs-whole-report half of it) is directly based on this paper.

GLoRIA

Huang, Shen, Lungren & Yeung, 2021, ICCV

This is probably the single most important paper for this project's design, so it got the most attention out of everything reviewed here. GLoRIA takes ConVIRT's idea and adds a second loss on top: instead of only matching "this whole image" to "this whole report", it also tries to match small patches of the image to individual words in the report, using an attention mechanism, without needing any extra bounding-box labels or an object detector. This "local" loss is what this project's Stage 1 is named after — "global InfoNCE + local patch-word loss" is literally GLoRIA's recipe. The local loss part hasn't actually been implemented yet (it currently exists as a stub in the code with a comment explaining why), but the global part that has been built is directly inspired by this paper together with ConVIRT.

BioViL

Boecking, Usuyama, Bannur et al., 2022, ECCV

This paper takes a different angle to get good results: instead of adding a local loss like GLoRIA, it puts most of the effort into making the text encoder itself better at understanding radiology language specifically — things like negation ("no evidence of pneumonia") and uncertainty ("cannot rule out"), which a generic language model tends to get wrong. Their custom text model is called CXR-BERT. This is a genuinely useful counterpoint to GLoRIA: it shows good grounding and zero-shot results are possible just from a better text encoder, without necessarily needing an explicit local contrastive loss at all. It's a reminder that this project's "local loss" choice is a design decision, not the only correct answer.

BiomedCLIP

Zhang, Xu, Usuyama et al., arXiv 2023, later NEJM AI 2025

This one matters very directly because it's the actual model this project's Stage 1 encoders are initialised from. BiomedCLIP is a CLIP-style model (ViT image encoder + PubMedBERT text encoder) trained on 15 million figure-caption pairs from biomedical papers on PubMed Central — not just X-rays, all kinds of medical images. Despite not being radiology-specific, it beat several specialist radiology models on various biomedical benchmarks. The bet being made here (and it is a bet, not a guarantee) is that starting from BiomedCLIP's weights instead of random weights or general CLIP gives Stage 1 a big head start, given that this project only has one small dataset and one GPU to fine-tune with, not the resources to train something like this from scratch.

3. Papers about classifying without training on labels (zero-shot)

CheXzero

Tiu, Talius, Patel, Langlotz, Ng & Rajpurkar, 2022, Nature Biomedical Engineering

This paper trained a CLIP-style model directly on ~377,000 MIMIC-CXR image-report pairs with absolutely no manual disease labels at all, and then did classification just by comparing an image's embedding against text prompts like "pneumonia" vs "no pneumonia" — matching board-certified radiologists on several conditions. It's a good preview of what could potentially be done with this project's own Stage 1 embeddings before Stage 2's classifier even exists: just prompt it with text and see how good the zero-shot predictions already are.

RadZero

Park, Yoon, Kim & Choi, arXiv 2025, NeurIPS 2025 poster

This is a very recent one (2025), so it's treated more as "an interesting recent direction" than "an established foundation" the way the older

papers here are. It proposes comparing text embeddings against local image regions directly (called VL-CABS) to get both zero-shot classification and pixel-level explainability maps out of the same contrastive representation, without a separate Grad-CAM step. It's worth including because it's pointing at basically the same combination this project is aiming for — alignment plus built-in explainability — just done in a more sophisticated way than fits within this project's scale.

4. Papers about generating reports

This is Stage 3 of the project, which hasn't been built yet (it's a later milestone), but it was read ahead of time to know what's involved before getting there.

R2Gen

Chen, Song, Chang & Wan, 2020, EMNLP

A Transformer-based report generator with some extra memory modules built in specifically to deal with the fact that radiology reports are very repetitive and templated (a lot of normal chest X-rays get almost the same sentence). What's useful here is that it was evaluated on IU X-Ray, which is the exact same dataset this project uses (also called Indiana/Open-i depending on where it's found). So this paper also serves as proof that this dataset is actually usable for report generation, not just classification.

LLaVA

Liu, Li, Wu & Lee, 2023, NeurIPS, oral

This is the architecture template Stage 3 of this project directly copies. The idea: take a frozen, already-trained CLIP image encoder, run an image through it to get patch features, project those features through one simple linear layer into the same "token space" that a language model understands, and then just feed them into the language model as if they were extra words at the start of the prompt. The language model itself (Vicuna, in their case) was fine-tuned on instruction-style data generated by GPT-4. This "frozen vision tower + linear projection + fine-tuned LLM"

pattern is exactly what this project plans to do with its own frozen Stage 1 vision encoder and a small LLM (Qwen2.5-0.5B-Instruct).

LLaVA-Med

Li, Wong, Zhang et al., 2023, NeurIPS Datasets and Benchmarks track (spotlight)

Basically LLaVA's recipe applied to biomedical images instead of general photos, using GPT-4 again to generate the instruction-tuning data, this time from PubMed Central figure captions. What stands out is that the whole thing was trained in under 15 hours on 8 GPUs — obviously still far more than a single GPU, but it's reassuring evidence that this general approach doesn't need an enormous compute budget to produce something reasonable, which is basically the bet the whole of Stage 3 here is making, just at an even smaller scale.

CheXagent

Chen, Varma, Xu et al., 2024, arXiv (technical report, not peer-reviewed as far as could be found)

This is the closest existing thing to what this whole project is trying to be — one system that does classification *and* generation *and* more, from a shared backbone. It was trained in four stages across 28 different public datasets covering classification, VQA, report generation, and grounding. It's a genuinely impressive system, but it's also roughly two orders of magnitude bigger than what a single student with one GPU can attempt, so it's treated here more as "the ceiling of what this idea can become" rather than something this project is trying to match.

5. Papers about making the model explain itself

Grad-CAM

Selvaraju, Cogswell, Das, Vedantam, Parikh & Batra, 2017, ICCV (extended version in IJCV 2020)

This is probably the most well-known explainability technique in all of deep learning. The idea is to take the gradient of a class score with respect to the feature maps of some convolutional layer, and use that to produce a rough heatmap of "which parts of the image mattered for this prediction", without changing the model or retraining anything. The plan is to use this on the classifier once it's trained on real labels, as a sanity check that it's actually looking at the lungs and not, say, a corner of the image with a hospital sticker on it.

Attention rollout

Abnar & Zuidema, 2020, ACL

This paper's main point is actually a warning: it shows that just looking at the last layer's raw attention weights in a Transformer is a bad, unreliable way to explain what the model is doing, because information gets mixed together across all the earlier layers first. Their fix is "attention rollout" — combining the attention weights across every layer, treated like a graph of how information flows through the network, to get a more honest picture. Since both the vision and text encoders in this project are Transformers (a ViT and a BERT-style model), Grad-CAM's usual convolution-based assumptions don't map onto them as cleanly, which makes this the more relevant explainability method for those parts specifically.

6. The paper about not needing a giant GPU

LoRA

Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang & Chen, arXiv 2021, published ICLR 2022

This is a very practical paper for this project. The problem it solves: fully fine-tuning a big pretrained model means updating every single weight, which needs a lot of GPU memory for the optimizer state, not just the model itself. LoRA's trick is to freeze all the original weights and instead train a pair of much smaller "low-rank" matrices that get added on top, which cuts the number of trainable parameters by a huge factor while still matching or beating full fine-tuning quality in their experiments, and adds

no extra cost at inference time once the update is merged back in. This is the only reason Stage 3's plan (fine-tuning a language model for report generation) is realistic on a single 16 GB GPU at all — full fine-tuning of even a "small" 0.5B model plus its optimizer state would be a tight squeeze otherwise.

So what's actually new here?

Honestly, not a new method — it's worth being upfront about that. GLoRIA already invented the local contrastive loss, LLaVA already invented the vision-to-LLM projection trick, LoRA already invented the efficient fine-tuning trick. What this project does is closer to *combining* these three separately-established ideas into one pipeline, at a scale that actually fits on a student budget: one GPU, one relatively small dataset, one semester. The systems that come closest to doing all three things at once (mainly CheXagent) do it with a much bigger model and a much bigger, multi-dataset training setup than is available here. The real question this project is asking is: how much of that combined behaviour can still be achieved if it's built small, on purpose, and instrumented with explainability all the way through instead of bolting that on at the end?

> The three papers this project keeps coming back to as main reference points are **GLoRIA** (for Stage 1's loss design), **CheXagent** (as the "what does the full version of this idea look like at scale" comparison), and **RadZero** (as a very recent example of someone else chasing the same alignment plus explainability combination this project is after).

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